

Implicit Modeling of Q-K-V Attention

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Abstract

Implicit deep learning models are used to jointly learn Q-K-V embeddings and contextual token representations designed to simultaneously capture language syntactic co-occurrence and semantic alignment. Four fundamental parameterizations elucidate the subtle and nuanced interplay between syntax and semantics.

Notation

$U = [t_1, t_2, \dots, t_N]$, an $r \times N$ input token matrix

$Y = [y_1, \dots, y_N]$, an $n \times N$ output matrix, $n > r$,

where n -vector y_i is a constant hyperparameter ϵ , drawn by Noken

Rademacher descent [3], or is given by a word embedding, like $\text{word2vec}(t_i)$

Implicit deep learning model [1]

$$X = \phi(KX + QU)$$

$$Y = VX + Q'U$$

where $K = T - \text{diag}(|T| \mathbf{1}_n) + \gamma \mathbf{I}_n$, $0 < \gamma < 1$, makes K well-posed for ϕ , that is fixed point X exists and is unique [2].

Shorthand, denote this implicit model by: $[K, Q, V, Q']$

Training problem

Over parameters (θ, K, Q, V, Q', X) minimize:

$$\{ L(Y, VX + Q'U) + \lambda_1 P_1(\alpha, Q', K) + \lambda_2 P_2(\theta, Q', K, V) \}$$

where $X = \phi(KX + QU)$

L is a loss function, P_1 and P_2 are penalty functions,
 $(\gamma, \alpha, \lambda_1, \lambda_2)$ are hyperparameters.

For mini-batch SGD, let batch size $1 < B < N$.

Let score be a similarity function, like a reproducing kernel. The degree to which word similarity captures meaning dictates the degree of query-key semantic alignment.

P1 attends to semantics and P2 attends to syntax as follows:

Let $k_i = Ks_i$, $q_i = Qt_i$, $v = Vs_i$, $q'_i = Q't_i$

where $s_i = i$ -th column of X .

For i in $[B]$, define probabilities $\alpha_{ij} = \text{softmax score}(q'_i, k_j)$,

a distribution of the i -th updated query q'_i across keys k_j .

Define penalty $P1(\alpha, Q', K) = (1/B) \text{Sum over } i, H(i, \alpha)$,

where $H(i, \alpha) = H_\alpha(\{\alpha_{ij}\})$, the average Renyi α -entropy of the α_{ij} .

Define h_i , contextual representation of token t_i , by:

$h_i = \text{Sum over } j, \alpha_{ij} v_j$, a probability weighted sum of values.

For L feedforward network layers, let $\theta = (W_L, \dots, W_1)$, where for $l \in [L]$,

W_l is a $n \times n$ weight matrix to be learned.

Let $h(0, i) = h_i$ and for l in $[L]$, define feedforward network update

$h(l, i) = \text{ReLu}(W_l h(l-1, i))$.

For i in $[B]$, define probabilities

$\beta_{ij} = \text{softmax score}(h(L, i), k_j)$,

a distribution of the L -th update $h(L, i)$ across keys k_j , and let $j(\beta, i) = \arg \max \beta_{ij}$.

Given input token t_i , define corresponding output token $o(i+1) = t_{j(\beta, i)}$.

Define penalty $P2(\theta, Q', K, V) = \text{Perplexity}[o_2, \dots, o(B+1)]$.

Parameterization Types (see diagram)

$[K, Q, V, Q']$

$[Q', Q, V, K]$

$[Q', Q, K, V]$

$[K, Q, Q', V]$

For the pairings (K, V) and (Q, Q') and distributions α of Q' over K , and contextual representations h , we just presented implicit model of attention with

parameterization 1. $[K, Q, V, Q']$. Modulo embedding input from U or X , we may similarly train

implicit models with parameterizations 2, 3, or 4. Each configuration has a different distribution α of Q' over K ; that is, different interaction between syntax and semantics.

Summary

Implicit deep learning models are used to jointly learn Q-K-V embeddings and contextual token representations designed to simultaneously capture language syntactic co-occurrence and semantic alignment. Four fundamental parameterizations elucidate different subtle and nuanced ways syntax and semantics can interact. They may be construed as different cognitive regimes: retrieval, prediction, abstraction, deliberation.

On each corpus chunk we use mini-batch SGD to jointly learn embeddings, fixed points and feedforward network parameters.

Given a prompt token, we look up the chunk it resides in and retrieve its corresponding pre-computed output token; an attention mechanism for both syntax and semantics, per parameterization.

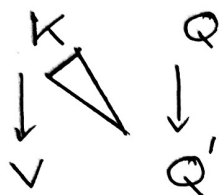
References

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Laurent El Ghaoui, Fangda Gu, Bertrand Travacca,
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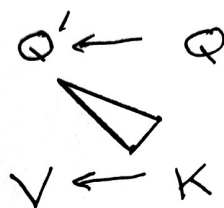
- [2] 'Robust Implicit Networks via Non-Euclidean Contractions'
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- [3] 'Noken Rademacher Descent'
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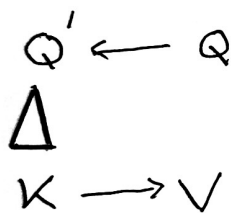
Parameterization diagrams



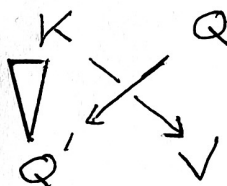
I



II



III



IV

Legend: $\rightarrow$ pairing

$\triangle$ distribution